



Guideline

Mupirocin Nasal Ointment for Decolonisation

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy

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Introduction

This guideline provides guidance to all Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOPHCG) staff involved in the management of perioperative or periprocedural *Staphylococcus aureus* decolonisation with nasal mupirocin. This document supports and complies with related legislation, policy and other related documents.

General information

Methicillin-resistant *Staphylococcus aureus* (MRSA) or Methicillin-susceptible *Staphylococcus aureus* (MSSA) carriage increases the risk of postoperative/procedural infection.

Staphylococcus aureus is commonly carried on the skin or in the nose of a large proportion of the population without causing illness or infection, however it can cause significant infection if it enters the body and into the bloodstream through broken skin e.g. cuts, surgical wounds or invasive devices including intravenous catheters.¹

Perioperative or peri-procedural decolonisation of colonised patients has been demonstrated to reduce the rate of postoperative/procedural infection in specific surgical procedures, in particular procedures involving the implantation of prosthetic material.

Therefore, eradication of *Staphylococcus aureus* carriage in pre-operative hospital patients is vital to prevent post-operative infection in high-risk patient groups.

Mupirocin is an effective topical therapy (often in combination with chlorhexidine or sodium hypochlorite wash) for the decolonisation of *Staphylococcus aureus* in colonised patients.¹ Dosage can depend on clinical setting and patient cohort.

Efficacy of decolonisation is optimised following administration of ≥ 6 doses of mupirocin. A randomised controlled trial assessing efficacy of mupirocin decolonisation demonstrated that 81% of *Staphylococcus aureus* carriers who received three to five doses of mupirocin were decolonised, however this rate increased to 93% when ≥ 6 doses were administered⁵.

The optimal timing of perioperative *Staphylococcus aureus* decolonisation is not known; ideally it should be started before the procedure however if this is not possible it can still be completed after the procedure.

This document is intended to provide standardisation and guidance to hospital staff on the administration of mupirocin nasal ointment for the decolonisation of *Staphylococcus aureus* carriage in specific high-risk populations, in accordance with the World Health recommendations².

Definitions

Decolonisation	A medical intervention aimed to reduce the microbial burden on an individual's body in an attempt to prevent transmission and potential infection.
Hickman™ line	A tunnelled large bore central venous catheter, inserted underneath the chest wall skin and intended for long term use
Non-tunneled central venous catheter	Non-tunneled central venous catheter (eg Vas-Cath™) insertion is performed following surgical aseptic technique, usually performed by an Anaesthetist in the designated areas of Post Anaesthetic Care Unit (PACU), Theatre, Radiology and Intensive Care Unit (ICU). With a non-tunneled catheter, the catheter enters the vein (eg internal jugular vein) near the skin puncture site (eg the catheter is inserted by puncturing directly into the skin and vein without tunnelling the subcutaneous tissue). They are intended for short term use (day to weeks only).
Perioperative	The time period including the preoperative, intraoperative and postoperative phases of surgery
Tunneled central venous catheter	Tunneled central venous catheter (eg Hickman™, Broviac™) insertion is performed following surgical aseptic technique and is usually performed in the Special Procedures area in Radiology. With a tunneled catheter, the catheter is tunneled under the skin subcutaneously before entering a large vein (commonly the internal jugular vein) to reach the heart. The skin exit site is distant from the venupuncture site to allow for optimal dressing and securement. They are designed for long term insertion (months to years).
Vascath™ line	A non-tunnelled large bore central venous catheter intended for short term use (generally used for temporary haemodialysis)

Indications

- **High risk patients requiring *Staphylococcus aureus* decolonisation**, including,
 - **Orthopaedic surgery** (procedures involving implantation of prosthetic material)
 - **Cardiothoracic surgery**
 - **Neurosurgery**
 - **Catheter insertion** (select catheter devices)
 - Central Venous Catheter (CVC) insertion
 - Non-tunnelled large bore catheters (e.g., Vas-Cath™)
 - Tunnelled (subcutaneous) large bore catheters (e.g., Hickman™, Broviac™, Groshong™)
 - Totally Implantable Venous Access Device (e.g Infusaport™, Port-a-cath™, PAS-Port™)
 - Peritoneal Dialysis (PD) catheter
- **MRSA decolonisation of patients or staff as directed by Infection Prevention and Control Unit.**

NB 1. The current available evidence does not support routine decolonisation for other non-high risk surgical procedures/interventions outside of the indications above.

For other specific high-risk clinical situations outside of the above in which the team would like to consider decolonisation therapy, please contact an Infectious Disease or Microbiology team member for advice.

NB 2. Within this document, decolonisation is not currently recommended for the routine insertion of short to medium term catheter insertions eg PICCs (peripherally inserted central catheter) or non-tunnelled, temporary (percutaneous) CVCs (central venous catheter).

Orthopaedic, Cardiothoracic and Neurosurgery

For orthopaedic surgery procedures involving implantation of prosthetic material and for all cardiothoracic and neurosurgery procedures

1. Apply nasal mupirocin 2% ointment inside BOTH nostrils TWICE a day for FIVE days (equivalent to a total of TEN doses), starting FIVE days prior to the operation.^{3,4} If unable to commence treatment FIVE days prior to operation then aim to commence as soon as possible within this 5-day period,

AND

Apply chlorhexidine 2% or 4% body wash ONCE daily for each of the FIVE days*

2. Once all recommended applications have been completed, discard tube and any remaining contents.
3. For any emergency surgery where the full decolonisation regimen cannot be completed prior to the operation each therapy (mupirocin ointment and chlorhexidine wash) should be initiated as soon as possible before surgery. This treatment may be continued following the procedure to complete the full FIVE-day decolonisation regimen.

*For cranial neurosurgical procedures it is important to ensure thorough application of the chlorhexidine wash to the hair and scalp for effective decolonisation

Catheter Insertion

For insertion of

- Non-tunnelled large bore central venous catheters e.g. Vascath™
- Tunnelled large bore central venous catheters e.g. Hickman™, Broviac™, Groshong™
- Totally Implantable Venous Access Device (e.g Infusaport™, Port-a-cath™
- Peritoneal dialysis (PD) catheter insertion.

NB. Decolonisation is not currently recommended for the routine insertion of short to medium term PICCs (peripherally inserted central catheter) or non-tunnelled, temporary CVCs (central venous catheter).

Pre-procedure

1. Apply nasal mupirocin 2% ointment inside BOTH nostrils TWICE a day for FIVE days (equivalent to a total of TEN doses), starting FIVE days prior to the procedure.^{3,4}
2. The mupirocin nasal ointment SHOULD also be accompanied by the application of chlorhexidine 2% or 4% body wash ONCE daily for each of the FIVE days.
3. For any emergency procedure where the full decolonisation regimen cannot be completed prior to the procedure, each therapy (mupirocin ointment and chlorhexidine

wash) should be initiated as soon as possible and continued up until the procedure (e.g., catheter insertion). This treatment may be continued following the procedure to complete the full FIVE-day decolonisation regimen (optional).

Post-procedure (maintenance therapy)

Applicable to

- Non-tunnelled large bore central venous catheters e.g. Vascath™
- Tunnelled large bore central venous catheters e.g. Hickman™, Broviac™, Groshong™
- Peritoneal dialysis (PD) catheter insertion.

NB. Maintenance application of mupirocin does NOT apply to patients who have had a Totally Implantable Venous Access Device insertion (eg Infusaport)

1. Apply nasal mupirocin 2% ointment inside BOTH nostrils ONCE a week following catheter insertion (to be continued for the duration that the catheter remains in situ)⁴

MRSA Decolonisation

For patients or staff (on advice of Infection Prevention and Control) requiring decolonisation of *Staphylococcus aureus*:

1. Application of nasal mupirocin 2% ointment TWICE daily inside BOTH nostrils for a total of FIVE days^{3, 4}
2. This should be accompanied by chlorhexidine 2% or 4% body wash for each of the FIVE days.
3. If chlorhexidine 2% or 4% body wash is not available or contraindicated (e.g., chlorhexidine allergy/intolerance) please contact an Infectious Diseases Physician, Infectious Diseases Pharmacist or Infection Control Officer for advice on a suitable alternative (e.g., dilute sodium hypochlorite solution ('bleach bath'))⁴.

Note 1: For mupirocin-resistant strains, nitrofurazone 0.2% nasal ointment may be an alternative option.

Prescribing Information

- Inpatient prescribing: prescribe mupirocin 2% nasal ointment and chlorhexidine body wash on the WA Hospital Medication Chart
- Outpatient prescribing: prescribe mupirocin 2% nasal ointment and chlorhexidine body wash on approved hospital prescription stationary.

Procedure to Apply Mupirocin Intranasally^{3,8}

1. Before administration, perform hand hygiene.
2. Squeeze a small dot of nasal ointment onto a cotton bud. The dot of ointment should be about the size of a match head.
3. Apply the ointment to the inside of each nostril.
4. Gently massage the outside of the nose with thumb and forefinger for 15 seconds to ensure the ointment is distributed around the nostrils.
5. Perform hand hygiene again once administration completed to minimise spread of infection.
6. Do not use any other nasal ointments or nasal sprays during the period of mupirocin treatment.

Adverse Effects^{3,8}

- Irritation, pruritus and stinging at the site of application³.

Practice Points³

- It is normal for patients to taste the mupirocin at the back of their throat post application if applied correctly.
- Do not mix with other products as this may dilute the drug concentration and impact overall antibacterial activity.
- Resistance may develop with widespread and prolonged use; limit duration to 5 days where possible.

Storage⁸

- Store below 25°C.
- After first opening the tube, the contents can be used for up to 7 days.

Related Documents**Legislation:**

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- MP007/18 Statewide Medicines Formulary Policy
- [OD0478/13 Infection Prevention and Control of Methicillin Resistant Staphylococcus aureus \(MRSA\) in Western Australian Healthcare Facilities \(HCFs\)](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGOPHCG Antimicrobial Stewardship Policy](#)
- SCGH HP141 Medicines Management Policy
- [OPH HWP Medicines Management Policy](#)
- [SCGH Medicines Management Procedures and Standards](#)
- [OPH Medicines Management: Procedures and Standards](#)
- NSQHS Standard 1.1.1: An organisation-wide management system is in place for the development, implementation and regular review of policies, procedures and/or protocols.
- NSQHS Standard 1.7: Developing and/or applying clinical guidelines or pathways that supported by best available evidence (all subsequent actions of this criterion are applicable)
- NSQHS Standard 4.1: Developing and implementing governance arrangements and organisational policies, procedures and/or protocols for medication safety, which are consistent with national and jurisdictional legislative requirements, policies and guidelines.
- NSQHS Standard 4.1.2: Policies, procedures and/or protocols are in place that are consistent with legislative, national, jurisdictional and professional guidelines.

Standards, procedures, guidelines:

- [SCGH Nursing Practice Guideline 33: Non-tunnelled and tunnelled large bore central venous catheters](#)
- [OPH Nursing & Midwifery Guideline Hickman lines \(tunnelled\) and large bore central venous catheters](#)
- [SCGH Nursing Practice Guideline 47: Infusaport – implantable intravenous delivery system](#)
- [SCGH Nursing Practice Guideline 51: Medication Management](#)
- [WA Health Public Health Information: Decolonisation treatment for MRSA](#)

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6. Large Bore Non-Tunnelled and Tunnelled Central Venous Catheters – Nursing Practice Guideline No 33. Sir Charles Gairdner Hospital
7. Infusaport – Nursing Practice Guideline No 47. Sir Charles Gairdner Hospital
8. Mupirocin Nasal Ointment (Medsurge). 2021. MIMS Australia

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Standards Applicable:	Relevant NSQHS Standards (Version 2):   				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form.

General guide for the application of antiseptic body wash

Important points

- a. Antiseptic body wash is intended for external use only. Avoid contact with your eyes, ears, mouth and other body orifices. *Check with your doctor or nurse before application if you have open wounds or broken skin present.*
- b. Do not swallow the solution.
- c. Use the antiseptic body wash undiluted.
- d. Before applying, remove all makeup, nail polish, jewellery and any piercings.
- e. Trim back fingernails and toenails to a short length and remove any false nails (if present).
- f. Wet the skin on your body prior to applying the antiseptic solution.
- g. When applying to your face, pay special attention to areas around the nose as microorganisms often colonise that area.
- h. Don't forget to scrub between the toes, fingers and soles of the feet.
- i. The chlorhexidine will bind to the skin and remain active despite being washed off.
- j. Do not apply any body creams, lotions, makeup, nail polish, piercings, hair products or deodorants following the antiseptic wash until after the procedure or surgery has been performed, or until advised by your health professional.



1 Wash your face with undiluted **HiBi SCRUB PLUS**, especially around the nose, avoiding your eyes and ears.



2 Using a clean sponge, working downwards, wash the armpits.



3 Continue washing downwards and wash well around the groin area.



4 After washing all these important areas, rinse off under the shower.



5 Now repeat steps 1 to 4, but this time starting with the hair.



6 After a final rinse, dry yourself thoroughly and put on clean clothes.